INTERNAL PRODUCT DOCUMENT · CONFIDENTIAL

TaskFlow

A clean, manual task manager. Users register, log in, and organise their own tasks with priorities, due dates, reminders, groups, and drag-and-drop ordering. A separate CRM admin panel lets the team manage users.

DOCUMENT TYPE
MVP / POC Product Requirements

VERSION
2.0 (supersedes v1.0)

STATUS
Final — scope locked

DATE
21 August 2026

PRODUCTS
TaskFlow App · TaskFlow CRM

AUDIENCE
Founder / engineering

STACK
Next.js + shadcn · NestJS · MongoDB · JWT

CHANGE FROM V1.0
AI agent removed from product

Scope is locked. There is **no AI feature inside the product**. AI is only a development tool used to build the app. Anything not listed as In-Scope must not be built in v1.

1. What we are building

TaskFlow is a straightforward to-do application with real accounts. A user signs up with their email, logs in, and manages their own tasks. Nothing is shared with other users, and all data lives in our own MongoDB database — no external services.

Alongside it, we ship a **CRM admin panel on a separate URL**. That is for us, not for customers. Admins log in with a fixed admin account and can view, create, edit, and delete users and their activity.

> **Two apps, one database:**
> **TaskFlow App** (`/`) — public sign-up, normal users, their own tasks.
> **TaskFlow CRM** (`/crm`) — admin login only, no registration, full visibility over users.

1.1 Important correction from v1.0

Version 1.0 described an AI agent that created and reorganised tasks. **That is removed.** Users create and manage everything manually. AI is used by us during development only, and is never part of the running product, the UI, or the database.

2. Document control

Field	Value
Document ID	TF-PRD-MVP-002
Version	2.0 — 21 Aug 2026 (replaces v1.0 entirely)
Owner	Product (founder)
Stack lock	Next.js App Router + Tailwind + shadcn/ui · NestJS · MongoDB / Mongoose · JWT in HttpOnly cookie
Repo shape	Turborepo: `apps/web` (app + CRM), `apps/api` (NestJS), `packages/ui`
Change policy	Scope changes require a version bump before code follows

3. Goals and non-goals

3.1 Goals

ID	Goal	Why
G1	Real accounts with JWT	Each user's tasks are private; JWT is mandatory
G2	Full manual task management	Create, edit, delete, complete, priority, due date
G3	Reminders on tasks	User sets a reminder time and is notified in-app

ID	Goal	Why
G4	Organise: groups, sort, drag-and-drop	Makes it a real to-do tool, not a flat list
G5	CRM admin panel	We can manage users without touching the database by hand
G6	Seeded demo users	We can log in and test immediately

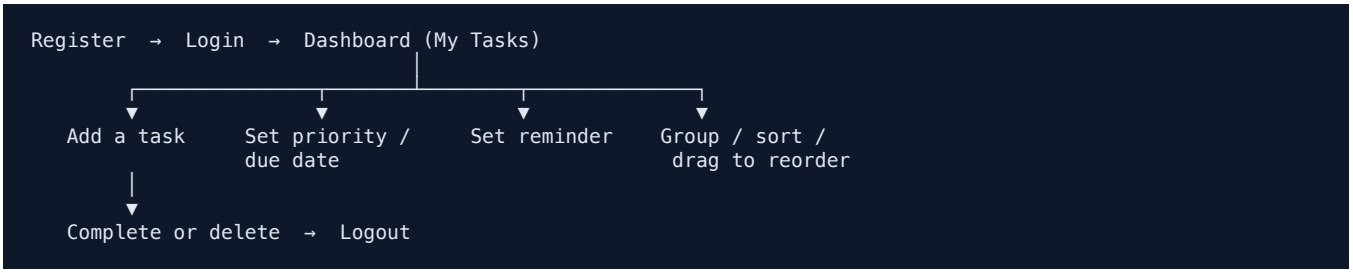
3.2 Non-goals (not in v1)

Out of scope	Reason
Any AI feature in the product	Removed by decision; users work manually
Task dependency graph / analytics dashboards	Was v1.0 scope; too heavy for this POC
Teams, sharing, comments	Single-user app
Email / SMS / push notifications	Reminders are in-app only for MVP
Calendar sync, mobile apps, billing, SSO	Not the wedge
Admin self-registration	CRM has login only; admins are seeded

4. Users

Who	Where	How they get access	What they can do
Normal user	TaskFlow App	Registers with their own email	Manage only their own tasks and groups
Demo user	TaskFlow App	Seeded by us	Same as normal user; used for testing
Admin	TaskFlow CRM	Seeded; no registration	Full visibility and control over all users

5. User journey (simple)



1. User opens the app, clicks Register, enters name, email, password.
2. Registration completes and they are logged straight in (JWT cookie set).
3. They land on My Tasks and add their first task.
4. They can set priority, a due date, and optionally a reminder time.
5. They organise tasks into groups, sort them, and drag to reorder.
6. They complete tasks (checkbox), edit them, or delete them.
7. Logout clears the session.

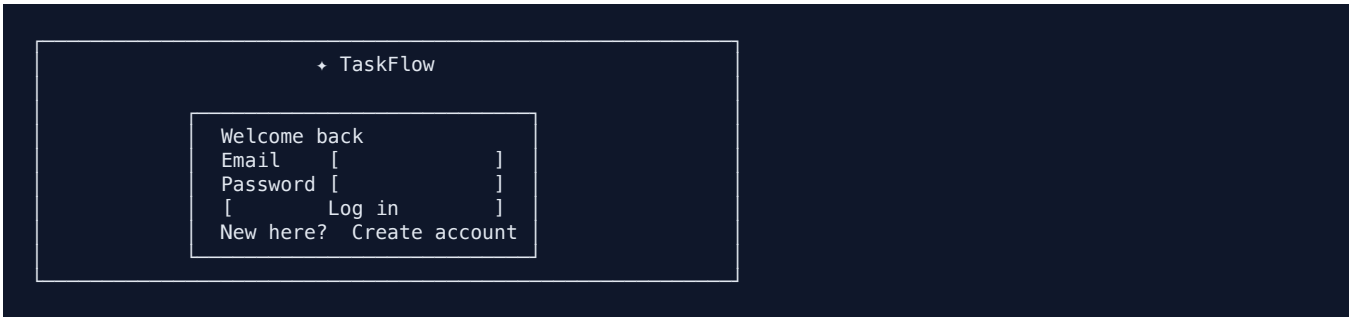
6. Feature scope — TaskFlow App

Feature	Behaviour
Register	Name, email, password. Email must be unique. Password hashed with bcrypt.
Login / Logout	Email + password. JWT issued in an HttpOnly cookie. Logout clears it.
Create task	Title (required), description, group, priority, due date, reminder.
Edit task	All fields editable inline or in a dialog.
Delete task	With a confirm step.
Complete / uncomplete	Checkbox toggle. Completed tasks move to the Completed filter.
Priority	Low · Medium · High, shown as a coloured badge.
Due date	Optional date. Overdue tasks are highlighted.
Reminder	Optional date + time. When it is due, the app shows a toast/banner while open. Can be edited or removed.
Groups	User-created lists in the sidebar (e.g. Work, Personal). A task belongs to one group or none.
Sort	By manual order, due date, priority, or created date.
Drag and drop	Reorder tasks within a list, and drag a task onto a group in the sidebar to move it.
Filters	All · Active · Completed. Plus Today and Overdue shortcuts.
Search	Simple title search within the user's tasks.
Profile	View name and email, change password, logout.

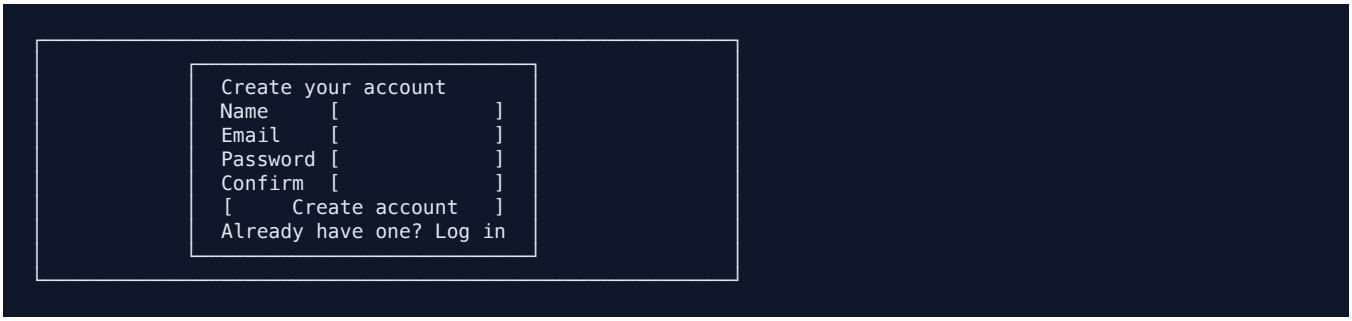
Reminder scope note: MVP reminders fire in the browser while the app is open (checked on load and on an interval). No email, push, or background worker in v1.

7. Screens and wireframes — TaskFlow App

7.1 Login

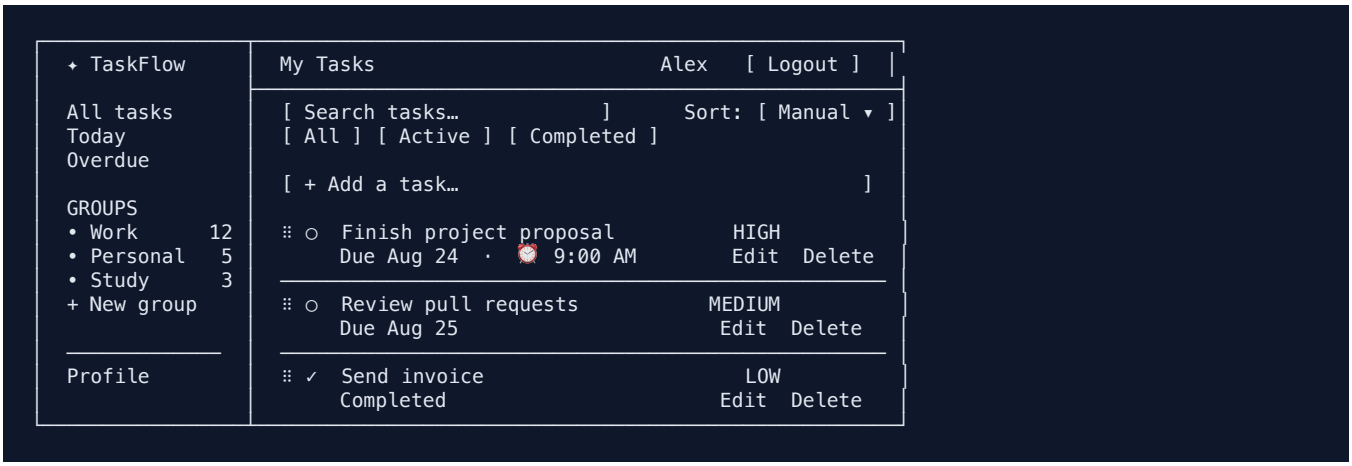


7.2 Register



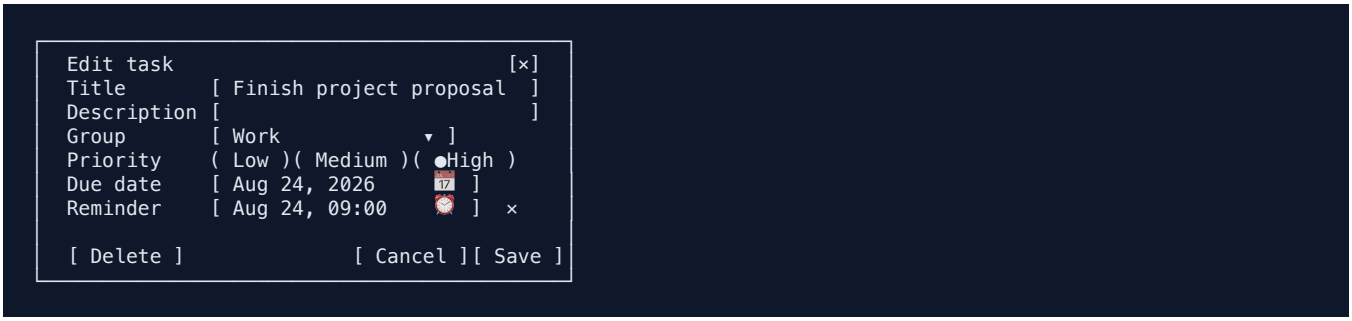
On success the user is logged in and redirected to My Tasks. Password minimum 8 characters.

7.3 My Tasks (main screen)



⠿ is the drag handle. ⏰ shows a reminder is set. Adding a task is a single inline input at the top — the fastest possible capture. Clicking a task opens the detail dialog.

7.4 Task detail / edit dialog



The × next to Reminder removes it. Everything here is optional except the title.

7.5 Profile

Shows name and email, a change-password form, and a Logout button. Nothing else in v1.

7.6 Empty states

- No tasks: “Nothing here yet. Add your first task above.”
- No groups: “Groups help you separate work and personal tasks. Create one.”
- Search with no result: “No tasks match that search.”

8. TaskFlow CRM (admin)

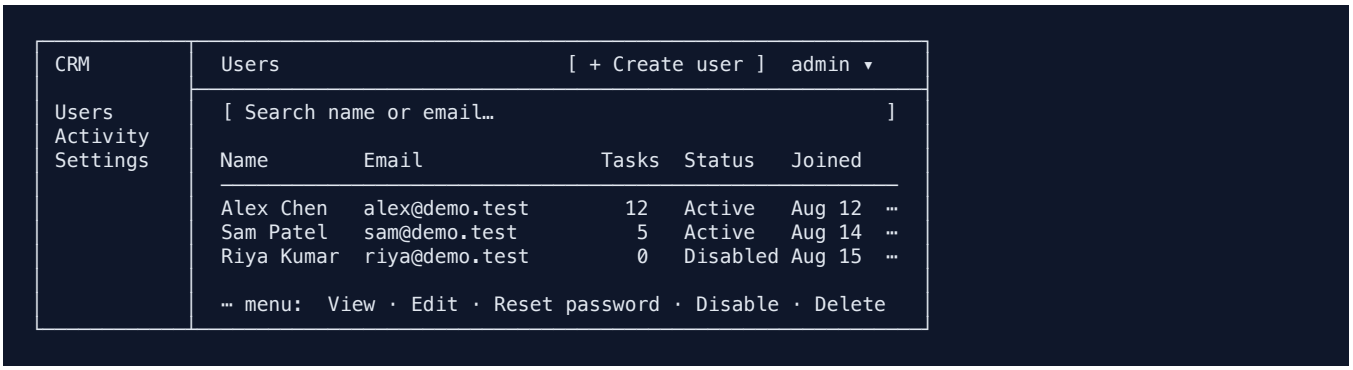
A separate area at `/crm`, with its own login and its own layout. It is not linked from the user app. There is **no registration** — admin accounts are seeded.

8.1 CRM login

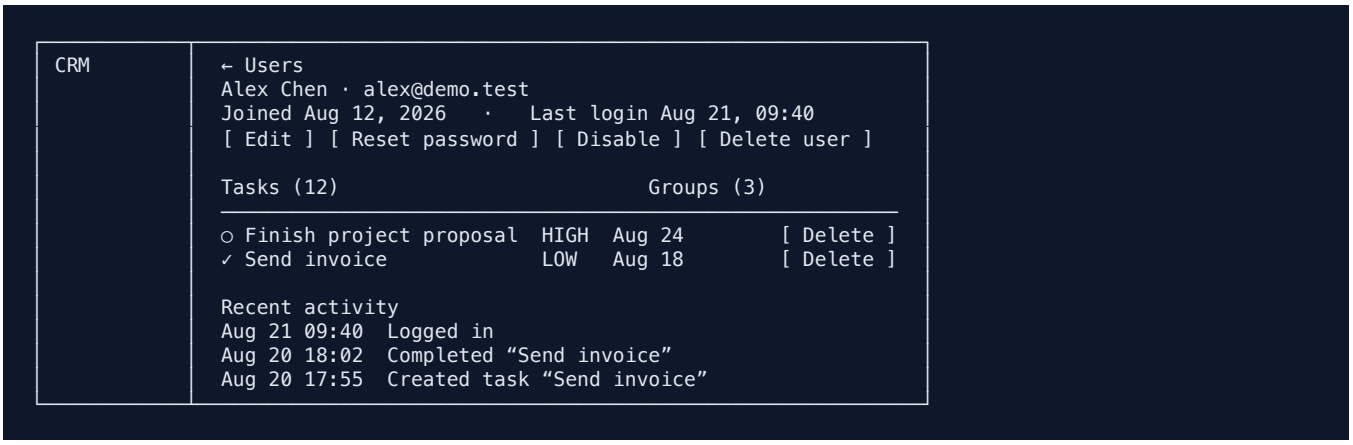


Admins log in with a username and password (not a public email sign-up). Separate JWT with an `admin` role; a normal user's token can never access CRM endpoints.

8.2 CRM users list



8.3 CRM user detail



8.4 CRM capabilities

Capability	Detail
List users	Search, sort, task counts, status, joined date
Create user	Admin manually creates an account (name, email, password) for testing or support

Capability	Detail
Edit user	Change name, email, status
Reset password	Set a new password for a user
Disable / enable	A disabled user cannot log in; data is kept
Delete user	Permanently removes the user and their tasks, with a typed confirmation
View tasks	See any user's tasks and delete individual ones
Activity log	Login, register, task created/completed/deleted events

Admin has full visibility and full privileges. Every destructive action is confirmed and written to the activity log.

9. Seeded accounts (for testing)

A seed script populates the database on first run so we can log in immediately. These are development credentials only and must be changed or removed before any public deployment.

Role	Login	Password	Seeded data
Admin (CRM)	admin	Admin@12345	Full CRM access
Demo user 1	alex@demo.test	Demo@12345	~12 tasks across 3 groups, some overdue, one reminder
Demo user 2	sam@demo.test	Demo@12345	~5 tasks, one group
Demo user 3	riya@demo.test	Demo@12345	Empty account, to test empty states

Anyone can still register with their own email — the seeded accounts are simply there so testing does not require sign-up first. All accounts live only in our MongoDB; nothing is sent anywhere else.

10. Data model (MongoDB)

10.1 User

Field	Type	Notes
name	string	required
email	string	required, unique, lowercase
password	string	bcrypt hash, never returned
status	enum	active | disabled
lastLoginAt	date	updated on login
createdAt / updatedAt	date	timestamps

10.2 Admin

Field	Type	Notes
username	string	required, unique
password	string	bcrypt hash
role	string	admin

10.3 Group

Field	Type	Notes
name	string	required
color	string	optional dot colour
order	number	sidebar position
userId	ObjectId	required, indexed

10.4 Task

Field	Type	Notes
title	string	required
description	string	optional
completed	boolean	default false
priority	enum	low | medium | high (default medium)
dueDate	date	optional
reminderAt	date	optional; null when removed
reminderFiredAt	date	set once shown, so it does not repeat
groupId	ObjectId	optional, ref Group
order	number	manual drag-and-drop position
userId	ObjectId	required, indexed
completedAt	date	set when completed
createdAt / updatedAt	date	timestamps

10.5 ActivityLog

Field	Type	Notes
userId	ObjectId	who it happened to
actor	string	user | admin

Field	Type	Notes
action	string	registered, logged_in, task_created, task_completed, task_deleted, user_deleted…
meta	object	e.g. task title
createdAt	date	indexed

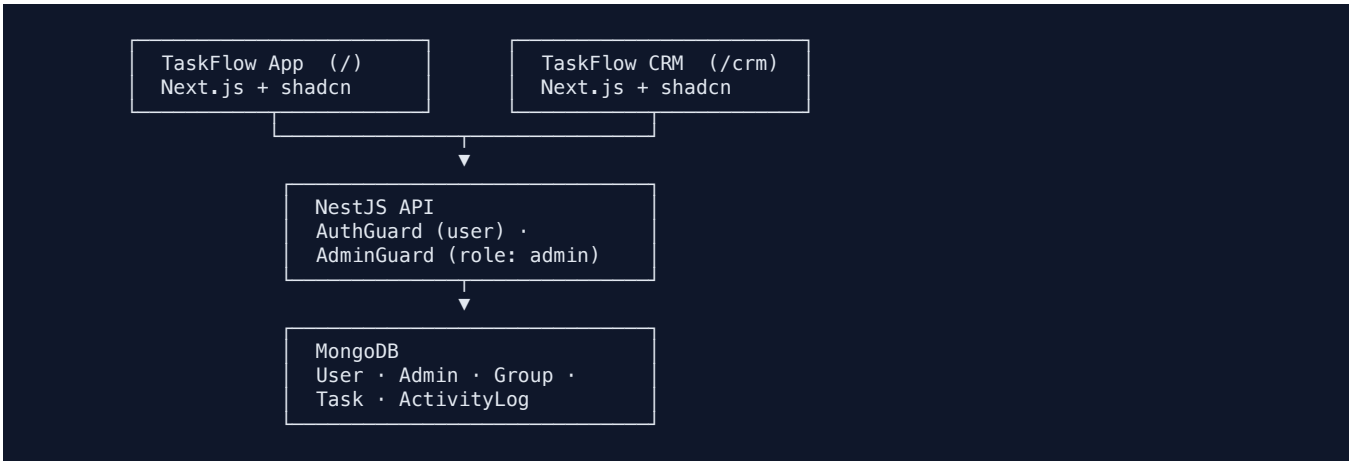
Indexes: unique `email` on User, unique `username` on Admin, Task `{ userId, order }`, `{ userId, completed }`, `{ userId, dueDate }`, `{ userId, reminderAt }`.

Tenancy rule: every task and group query filters by `userId` taken from the JWT. The client never sends a trusted user id. Only admin endpoints may query across users.

11. API

Method	Endpoint	Purpose
POST	/api/auth/register	Create account, set JWT cookie
POST	/api/auth/login	Log in, set JWT cookie
POST	/api/auth/logout	Clear cookie
GET	/api/auth/me	Current user
GET / POST	/api/tasks	List (filter, sort, search) / create
PATCH / DELETE	/api/tasks/:id	Update (incl. complete, reminder) / delete
PATCH	/api/tasks/reorder	Persist drag-and-drop order
GET / POST	/api/groups	List / create group
PATCH / DELETE	/api/groups/:id	Rename / delete (tasks become ungrouped)
POST	/api/admin/login	Admin sign in (separate token, role admin)
GET / POST	/api/admin/users	List users / create user
GET / PATCH / DELETE	/api/admin/users/:id	Detail / edit or disable / delete
POST	/api/admin/users/:id/password	Reset a user's password
GET	/api/admin/activity	Activity log

12. Architecture



Layer	Choice	Reason
Monorepo	Turborepo + pnpm	One repo for web and api
Frontend	Next.js App Router	Route groups keep app and CRM layouts separate
UI	Tailwind + shadcn/ui	Clean, consistent, fast to build
Drag and drop	dnd-kit	Accessible, works well with lists
Backend	NestJS	Modules and guards keep user vs admin access clean
Database	MongoDB + Mongoose	Simple documents, everything in our own DB
Auth	JWT in HttpOnly cookie + bcrypt	Mandatory JWT, token never exposed to JavaScript

Nest modules: `AuthModule`, `UsersModule`, `TasksModule`, `GroupsModule`, `AdminModule`, `ActivityModule`, `SeedModule`.

13. Security

Control	Implementation
Passwords	bcrypt, cost 10–12, never returned by the API
Session	JWT in HttpOnly, SameSite=Lax cookie; Secure in production
Roles	Separate user and admin tokens; AdminGuard on all `/api/admin/*`
Tenancy	Every user query scoped by `userId` from the token
Login errors	Generic “Invalid credentials” — no account enumeration
Validation	class-validator DTOs with a global ValidationPipe
Rate limiting	On login and register endpoints
Secrets	`MONGO_URI`, `JWT_SECRET`, `ADMIN_SEED_PASSWORD` in env, never committed

Control	Implementation
Disabled users	Blocked at login and rejected by the guard

14. Look and feel

Aspect	Decision
Style	Clean and light, neutral slate palette, one accent colour
Priority colours	High = red badge, Medium = neutral badge, Low = outline badge
Density	Comfortable list rows, clear tap targets, keyboard friendly
Feedback	Optimistic updates on complete and reorder, toasts for save/delete, skeletons while loading
Motion	Short 150–200ms transitions; drag feels responsive, nothing decorative
CRM	Same components, denser table layout, clearly labelled as admin
Responsive	Sidebar collapses to a drawer on mobile

Components used: Button, Input, Textarea, Card, Badge, Checkbox, Tabs, Select, Dialog, Dropdown Menu, Sheet, Table, Calendar, Skeleton, Sonner (toasts), Sidebar.

15. Build plan

Phase	Deliverable	Done when
0 · Setup	Turborepo, Next + shadcn, NestJS, MongoDB connection, env files	`pnpm dev` runs web and api together
1 · Auth	Register, login, logout, me, JWT cookie, route protection	Two users log in separately and stay logged in after refresh
2 · Tasks	CRUD, complete, priority, due date, filters, search	A user can run a full task lifecycle
3 · Organise	Groups, sort options, drag-and-drop ordering	Order and group changes survive a refresh
4 · Reminders	Reminder field, in-app notification, edit/remove	A due reminder appears once and does not repeat
5 · CRM	Admin login, users list, detail, create/edit/disable/delete, activity log	Admin manages a user end to end
6 · Seed & polish	Seed script, empty states, errors, README	Definition of Done is fully green

16. Definition of Done

1. A visitor can register with their own email and is logged in immediately.

2. Login, logout, and session-after-refresh all work using a JWT cookie.
3. Two different users never see each other's tasks or groups.
4. Tasks support create, edit, delete, complete/uncomplete, priority, and due date.
5. Reminders can be set, edited, removed, and fire once in the app.
6. Groups can be created and renamed; tasks can be moved between them.
7. Tasks can be sorted and reordered by drag and drop, and the order persists.
8. Filters (All, Active, Completed, Today, Overdue) and search work.
9. CRM at `/crm` has admin-only login with no registration.
0. CRM can list, create, edit, disable, and delete users, and view their tasks and activity.
1. Seed script creates one admin and three demo users with sample data.
2. README documents environment variables, how to run, and the demo credentials.
3. No AI feature exists anywhere in the running product.

17. Open points

Question	Default if undecided
Should deleting a group delete its tasks?	No — tasks become ungrouped
Do reminders need to work when the app is closed?	No in v1; revisit with a worker or web push later
Should users be able to delete their own account?	Not in v1; admin can delete from CRM
More than one admin?	Seed one; the schema supports more

End of document — TF-PRD-MVP-002 v2.0 · TaskFlow · 21 August 2026